Anuj Pratap Singh Yadav

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SUMMARY

Staff engineer with 11 years of C++ systems at Synopsys, Microsoft and Adobe. I own the coverage subsystems of a formal-verification product, lead cross-team architecture changes, and build LLVM program analysis and AI agents for the team.

SKILLS

Languages: C++, Python, Tcl, C#, Java, TypeScript

Domain: VC Formal (DPV, FPV), formal verification, C++ and RTL code coverage, cone of influence, property density

Systems and tools: dependency inversion, IWYU, LLVM IR passes (dg), master/worker orchestration, XML (pugixml), Verdi, LSF, GDB, Linux, Perforce, Git, Win32

AI tooling: Cursor agents, skills, rules and subagents

EXPERIENCE

R&D Staff Engineer | Synopsys *Noida, India | May 2023 - Present*

VC Formal (formal verification), DPV app: C++ vs RTL equivalence

- **Build-dependency decoupling (multi-org initiative)**: led VC Formal's part. The shared C++ layer under ~15 apps and tools included engine headers, so one engine edit rebuilt every app. Cut wrong-way dependencies ~73% with include-what-you-use, hide-what-you-use (forward declarations), dependency inversion through startup-injected callbacks, and code relocation, in small changes reviewed by the owning teams; in production. Org-wide tech talk on the design.
- **C++ code coverage**: own it end to end. It shows which parts of a C++ model the formal proofs actually exercised, so C++ can be signed off like RTL. Added merging across proofs and export to the standard coverage database, so C++ and RTL coverage are reviewed together in one viewer; handle customer issues and guided its integration into a new C++ + RTL sign-off app.
- **C++ program slicer (customer-requested R&D)**: building source-level cone of influence for C++ designs, which reports the exact C++ lines a property depends on. Built on the open-source dg LLVM slicer: wrote LLVM IR passes for C++ constructs it missed and a signal-based slicing criterion that cut whole-file slices down to the lines feeding each signal. Design reviewed weekly with a Distinguished Engineer.
- **Formal coverage (property density, RTL coverage)**: own both. They report which design signals, down to the bit, and which RTL lines the properties actually check. Rebuilt property density on the shared design model so DPV and FPV give the same answer, adding bit-level detail and RTL-to-RTL designs; aligned RTL coverage semantics across the two apps. In production.
- **Licensing for a distributed solver**: built token licensing in the C++ proof engine and moved runtime license checkout from workers to the orchestration master, so each key serves more workers; fixed production checkout failures across license servers.
- **Customer escalations**: own them end to end in my areas: root cause, regression test, fix shipped in service packs.
- **AI agents for the team's workflow**: built Cursor agents teammates use: a daily Jira check that flags tickets with wrong validation fields and notifies owners, and an agent that reproduces a Jira issue, root-causes it and drafts a fix as a code review for an engineer to check.
- **Session save/restore**: closed silent state-loss gaps (user Tcl procedures, engine state as XML), with round-trip tests.

Software Engineer II | Microsoft *Noida, India | May 2021 - May 2023*

- Owned Recording Studio features in PowerPoint Win32 (C++): camera modes, live background blur, video export and Cameo, a camera object placed on the slide. Led v1 test automation and wired Recording and Export telemetry.
- Designed catalog filtering plus an allowlist (C#) so new commerce SKUs no longer broke OneDrive's upgrade path.
- Built the Stream 2.0 video playback component in Excel Win32 (C++, TypeScript) for SharePoint and OneDrive video.

Software Development Engineer 2 | Adobe *Noida, India | Aug 2015 - Apr 2021*

- Moved Photoshop and Premiere Elements from RIBS to Hyperdrive builds with parallel bundles, cutting install time ~50%.
- Designed lazy loading for large files in the AEM web editor, cutting load time ~60% on files customers reported as freezing.

EDUCATION AND AWARDS

Spot Awards, Synopsys (latest Sep 2026, org all-hands) | Spot Award, Adobe (RIBS to Hyperdrive migration)

B.Tech, Computer Science and Engineering - NIT Bhopal (MANIT) | 2011 - 2015 | GPA 8.27/10